Xinyi Wang

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Research Interests:.LLM-driven Human-Robot Interaction, Embodied AI, Interactive Agents, Robot Learning, Multimodal Systems

EDUCATION BACKGROUNI

Xi'an Jiaotong University (XJTU), *Xi'an* *Sep.2017-Jul.2021*

- BS in Energy and Power Engineering A; GPA:85.23/100

Peter the Great Saint-Petersburg Polytechnic University, *St. Petersburg* *Aug.5-Sep. 5, 2019*

- International Polytechnic Summer School Program "Digital Engineering"

Northwestern University (NU), *Evanston* *Sep.2021-Jul. 2022*

- MS in Robotics and Control; GPA:3.643/4.0

Purdue University (PU), *West Lafayette* *Sep.2023-Present*

- Phd in College of Engineering

PUBLICATIONS

Towards Design Guidelines for Natural-Language-Based Human-Multi-Robot Collaboration in Domestic Environments

Xinyi Wang*, Shao-Kang Hsia, Ziyi Liu, Chenfei Zhu, Zhengzhe Zhu, Xiyun Hu, Anastasia Kouvaras Ostrowski, Karthik Ramani*

ACM CHI 2026 Poster(EA), Accepted

- Derived design guidelines for natural-language-based human–multi-robot collaboration in domestic environments
- Conducted a VR role-playing study to analyze the effects of autonomy, coordination dominance, and robot personality on user experience

Towards Natural Language Environment: Understanding Seamless Natural-Language-Based Human–Multi-Robot Interactions

Ziyi Liu,* ***Xinyi Wang******,** *Shao-Kang Hsia*, Chenfei Zhu, Zhengzhe Zhu, Xiyun Hu, Anastasia Kouvaras Ostrowski, Karthik Ramani*

arXiv preprint, 2026

- Introduced the concept of Natural Language Environment (NLE) for LLM-based human–multi-robot interaction.
- Conducted a VR role-playing study to derive a design space spanning autonomy, task coordination dominance, and robot personality.

AgentCoach: LLM-Based Adaptive Coaching Feedback for Motor Skill Learning

Dizhi Ma, Jiakun Yu*,* ***Xinyi Wang****, Xiyun Hu, Liang He, Sooyeon Jeong, Karthik Ramani*

ACM CHI 2026, Accepted

- Proposed an LLM-powered coaching agent that extracts coaching points from tutorial videos and maps high-level semantics to kinematic parameters.
- Designed adaptive feedback strategies that evolve with learner performance history, mimicking expert human coaches.

JustShape: Exploring Co-Speech Gestures for Multimodal LLM-Empowered Generative Parametric Modeling

Runlin Duan, Yuzhao Chen, Yichen Hu, Ziyi Liu, Chenfei Zhu, Xiyun Hu, Dizhi Ma, ***Xinyi Wang****, Karthik Ramani*

ACM CHI 2026, Accepted, Honorable Mention Award

- Investigated co-speech gesture as a complementary modality to language for expressing 3D design intent.
- Developed a multimodal fusion pipeline that translates speech and gestures into parametric modeling constraints.

Optimized Automated Suture Needle Segmentation and Re-grasping Using the Da Vinci System

Xinyi Wang*, Richard Voyles, Juan Wachs*

ICRA@40 Extended Abstract, 2024

- Presented an automated vision-based needle segmentation and re-grasping pipeline on the Da Vinci surgical robot.

PROFESSIONAL EXPERIENCE

Research

LLM-Based Interactive Agents for Skill Learning (AgentCoach)

- Built an end-to-end system that converts instructional videos into actionable coaching feedback using multimodal LLMs.
- Designed a coaching-point–centric evaluation framework linking language instructions to measurable pose parameters.
- Conducted controlled user studies comparing adaptive vs. non-adaptive feedback strategies.

Human–Multi-Robot Interaction with Natural Language

- Explored how users conceptualize and coordinate multiple heterogeneous robots using natural language.
- Synthesized prior HRI literature into a structured design space and validated it through VR-based role-playing studies.
- Implemented Unity + VR environments for role-playing studies, enabling rapid prototyping of multi-robot interaction scenarios.

Automated Suture Tasks on Da Vinci Surgical System, *Purdue University*

- Integration and optimization of the YOLO (You Only Look Once) model, which is trained on a self-collected dataset for needle tracking during the suturing procedure;
- A successful application of reinforcement learning on the Da Vinci system, utilizing both binary and dense reward functions based on suture status;
- Demonstration through experiments that the computer vision model, reinforcement learning-based motion planning model can work; (iii),a suture evaluation model and metrics to assess quality suturing outcomes.

Measuring and reducing the error of Visiflex Sensor

- Mount the visiflex to FT sensor and collect pose of visiflex dome and force torque data from both side at same time with Jetson. And validate the PnP solutions and calibrate the camera used in visiflex.
- Optimize the visiflex hardware, mounting method and code for collecting data to reduce the error.

  Tools: **MATLAB, ROS, OpenCV, Python, C++.**

Fabricating Flexible Exoskeletons for Robots

- Aimed to use solenoid valve, SCM and vacuum pump to realize the function of inhalation, use gyroscope to measure the angle of steam dome, and use pressure sensor to measure pressure.
- Assembled, collected and formulated data curves, and then made comparisons with curves of other models to seek for the best possible optimizations.

  Tools: **Arduino, C++, AutoCAD.**

Development of a Lower Limb Exoskeleton Robot

- Aimed to design a hip-knee two-DOF controller lower limb exoskeleton robot.
- Clarified the research background, overseas and domestic research status, key techniques to be solved, modeling, simulation and assembling involved in this topic.

  Tools: **Solid Works.**

RESEARCH FOCUS

I study how large language models can be integrated with perception and control to create interactive, human-centered robotic agents capable of learning skills, adapting behavior, and collaborating with humans.

Projects

Robotics Manipulation Projects, *Northwestern University*

- Analyze the Kinematics of a Multi-degree of freedom mobile robotic vehicle (a Kuka youBot manipulator mounted on a chassis with 4 wheels) and calculate a proper trajectory for garbing something up.
- Have the robotic vehicle follow a pre-defined trajectory to grab a block, which has located in a changeable place in simulation system, and put it in a new location.

  Tools: **Python, CoppeliaSim.**

EKF Simultaneous localization and mapping Projects, *Northwestern University*

- Analyze the forward and inverse kinematics of turtlebot and move it in both rviz and real world by using keyboard and ROS command. And amending the path of robot using EKF algorithm.
- Compare and display the path of turtlebot in real world, odometry and EKF slam.

  Tools: **ROS, C++, Python, URDF, Kalman Filter.**

Avatar Robot Design, *Northwestern University*

- Create a two-degree-of-freedom interactor robot (the avatar) and a corresponding haptic display robot (the interface). The interface can be served as a remote controller for the avatar, which defines a fingertip haptic display when the coupling avatar's fingertip touches across a surface which various in height and texture.

  Tools: **Python, C++, SolidWorks.**

WORKING EXPERIENCE

Robotics Software Engineer in Robotics and Control team Company: Johnson&Johnson.

- Write the auto-assignment algorithm to let Ottava Combo system's 3 robot arms assign to left and right based on their relative position with endoscope during tele-operation, which is on one robot arm.
- Build a new server and client structure by using Grpc to let the platform communicate with GUI. Give GUI the assignment result to let GUI show it on screen and Get the confirm information from GUI.

  Tools: **gRpc, C++.**

SKILLS

Software (Language) Skills

- AutoCAD, Origin, Solid Works, Fluent, MATLAB, Fortran, C, C++, Python, OpenCV, Grpc, Unity (simulation, XR, HRI prototyping)

Machine Learning

- Nearest Neighbor Classifiers, Linear and Polynomial Regression, Decision Trees, Gradient Descent, Clustering, Reinforcement Learning, Multilayer Perceptron, Convolutional Networks.